Noise sensitivity — output quality vs input noise

White Gaussian noise added to the input before separation, at input-SNR levels clean / 20 dB / 10 dB / 0 dB. FEC held fixed at 10% burst loss, staggered placement. Scores are vs the clean-input separation, so they fold together separation error, input noise, and packet loss. All scores 0-100, higher = better.

lamour de ma vie

16,725 packets · 0.4% silent

Noise	Mode	SDR-V	SDR-I	CDPAM	ViSQOL
clean	EEP	41.4	44.0	95.1	90.0
	UEP	30.2	33.5	91.8	84.3
20 dB	EEP	33.6	27.2	48.3	11.4
	UEP	27.1	24.0	48.9	12.3
10 dB	EEP	25.0	0.0	39.3	10.9
	UEP	21.2	0.0	39.6	12.0
0 dB	EEP	0.0	0.0	28.4	11.7
	UEP	0.0	0.0	28.4	12.7

man i need

10,075 packets · 9.6% silent — most noise-robust

Noise	Mode	SDR-V	SDR-I	CDPAM	ViSQOL
clean	EEP	37.8	35.0	94.5	93.1
	UEP	46.1	36.9	96.0	93.3
20 dB	EEP	36.7	33.9	83.5	42.2
	UEP	45.2	35.6	83.3	42.4
10 dB	EEP	31.9	25.8	74.1	11.6
	UEP	38.1	26.3	74.3	11.4
0 dB	EEP	17.3	2.9	64.4	2.1
	UEP	16.5	4.3	64.0	2.1

the cure

14,750 packets · 0.1% silent

Noise	Mode	SDR-V	SDR-I	CDPAM	ViSQOL
clean	EEP	41.9	38.1	94.0	92.0
	UEP	36.9	34.5	93.3	89.0
20 dB	EEP	39.7	32.2	60.1	20.7
	UEP	35.9	30.1	59.4	19.6
10 dB	EEP	27.7	10.0	49.1	9.5
	UEP	25.7	9.2	48.7	9.5
0 dB	EEP	16.7	0.0	41.1	7.0
	UEP	15.9	0.0	40.8	6.9

SDR-V / SDR-I = SI-SDR on the separated streams · CDPAM / ViSQOL = perceptual, on the recombined song. UEP rows in blue.
Key finding: quality is dominated by separation corruption under noise — the EEP/UEP gap shrinks to near zero once noise (not

packet loss) becomes the bottleneck. ViSQOL is far more noise-sensitive than CDPAM.